ALGO

Algae Biorefinery Project

FOOD BIOTECHNOLOGY

Course Code BTH 638

PRESENTED BY

Ifrah Naseem	B22106006022
Sana Naz	B22106006048
Zainab Irfan	B22106006063

ALGAE BIOREFINERY PROJECT

Meet ALGO

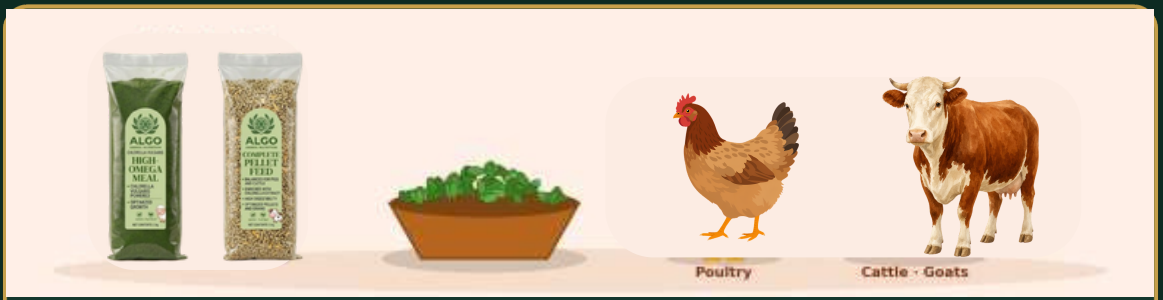
ALGO is an integrated microalgal biorefinery concept instead of treating algae as one raw material,we match each strain's natural biology to the product it suits best,then tailor cultivation,harvesting and processing to it.

ONE PLATFORM, MULTIPLE PRODUCTS.



01 · ALGAE OIL

A nutritional oil from an oil-producing microalga, rich in omega-3 fatty acids such as DHA or EPA. Offered in three flavours: classic, chili and mushroom.



02 · ALGAE ANIMAL FEED

Protein-richalgalbiomass,includingthe biomass left after oil extraction, as a feed ingredient for poultry, cattle and goats.



03 · ALGAE BIOSTIMULANT

Asprаymade from algalextract tosupport seedling growth, root development and nutrient-use efficiency in plants.

Benefits are goals to be validated through testing.

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Methodology

From strain selection to three products

STEP 1 · STRAIN SELECTION

Schizochytrium sp.

For algae oil (DHA)

Heterotrophicmarine protist grown on organiccarbon. A well-studied DHA producer; now also classed as Aurantiochytrium. [1][2]

Nannochloropsis sp.

For EPA-rich oil option

Photosyntheticmarinemicroalga known for accumulating EPA.EPAisnotDHA,soitsuits a different oil. [3]

Chlorella vulgaris

For feed and biostimulant

Protein-richgreenmicroalgastudied as animal feed and as a plant biostimulant. [4][5][6][7][8]

↓ each strain follows its own route ↓

ALGAE OIL	ANIMAL FEED	BIOSTIMULANT
Fermentation Schizochytriumsp. grown on organic carbon	**Cultivation** Chlorellavulgaris in a modular photobioreactor	**Cultivation** Chlorellavulgaris in a modular photobioreactor
↓	↓	↓
Biomass recovery Separatethecellsfrom the broth	**Harvesting** Filtrationoranother low-cost separation	**Harvesting** Separatebiomass from the culture
↓	↓	↓
Cell disruption Openthecellstorelease the lipids	**Cell-wall disruption** Improvenutrientavailability for animals	**Extract preparation** Biomassextractorcell-free culture liquid
↓	↓	↓
Extraction and purification Recover andrefinethe DHA-richoil	**Drying and processing** Biomass,plusresidualbiomass from oil	**Formulation** Spray-ready,tested at several concentrations
↓	↓	↓
Quality and oxidation testing Safety, shelflifeandflavourstability	**Feed evaluation** Composition,digestibility, safety, inclusion level	**Seedling trial** Treatedvsuntreated, identical conditions
↓	↓	↓
Algae oil classic · chili · mushroom	Algae feed ingredient poultry · cattle · goats	Algae biostimulant spray seedlings · gardens · farms

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References

Research behind the strains and applications used in ALGO

ALGAE OIL: SCHIZOCHYTRIUM (DHA)

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EPA OPTION: NANNOCHLOROPSIS

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ANIMAL FEED: CHLORELLA VULGARIS

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BIOSTIMULANT: CHLORELLA VULGARIS

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8. La Bella, E., Baglieri, A., Rovetto, E. I., Stevanato, P., & Puglisi, I. (2021). Foliar spray application of Chlorella vulgaris extract: Effect on the growth of lettuce seedlings. Agronomy, 11(2), Article 308. https://doi.org/10.3390/agronomy11020308